FOR THE TUTOR OR PARENT

Answer key: Ages 8–10

Accept equivalent forms (for example 70 cents and 70c, or 3/4 and three quarters). One mark per question. Add the warm-up and main marks to get a score out of 10; the Brain stretchers are a bonus and are not counted. Keep this key away from the child until they have finished.

Warm-up (4 marks)

#	Question	Answer	Watch for
1	47 + 38 = ?	**85**	Watch for forgetting to carry the 1 (gives 75).
2	305 - 128 = ?	**177**	Watch for borrowing across the zero; common error 223 (smaller digit minus bigger).
3	There are 4 packets. Each packet has 8 stickers. How many stickers in all?	**32**	Does the child know 8 x 4 or count/add in 8s? Note speed and strategy.
4	24 sweets are shared equally among 3 children. How many sweets does each child get?	**8**	Check the child shares equally (division), not subtracts 24 - 3.

Main questions (6 marks)

#	Question	Answer	Watch for
5	Find the missing digit: 2 _ 7 + 358 = 645. What digit goes in the blank?	**8**	Ask how they found it. Good: work from the ones (7 + 8 = 15, carry 1). Guess and check is fine; note whether the child started from the ones column.
6	Mei has 3 times as many stickers as Ravi. Together they have 48 stickers. How many stickers does Mei have?	**36**	Look for a bar model: Ravi 1 unit, Mei 3 units, 4 units = 48, 1 unit = 12. Common error: 16 (that is 48 / 3) or 12 (Ravi's).
7	52 children sit in rows of 7. How many children are left over?	**3**	Child should use 7 x 7 = 49, then 52 - 49 = 3. Accept 7 r 3 or 7 rows and 3 left over, but the answer to the question asked is 3. Common error: 7 (the number of rows) or 4 (from rows of 8). Note whether the child knows the 7 times table or counts on in 7s.
8	Fill in the blank: 3/4 = ?/12	**9**	Does the child multiply top and bottom by 3? Common error: adding 8 to both (11/12).
9	A rectangle is 9 cm long and 4 cm wide. What is its area?	**36 (any sensible unit: 36 cm2, 36 sq cm or 36 square cm)**	Award the mark for 36 even if the unit is missing or is cm only, and note it for the tutor chat. Common error: giving the perimeter, 26. Ask what area means.
10	A film starts at 2:45 p.m. and lasts 1 hour 50 minutes. At what time does it end?	**4:35 p.m.**	Look for jumping: +1 hour = 3:45, +15 min = 4:00, +35 min = 4:35. Common error: 3:95 or 4:15.

Brain stretchers (bonus, not counted)

#	Question	Answer	Watch for
11	I think of a number. I multiply it by 6, then subtract 4. My answer is 38. What is my number?	**7**	Working backwards: 38 + 4 = 42, 42 / 6 = 7. Guess and check is also fine.
12	Use the digits 2, 5, 8 and 0 to make 2-digit numbers. Each number must use two different digits, and a number cannot start with 0. How many different numbers can you make?	**9**	Are they listing in order (20, 25, 28, 50, 52, 58, 80, 82, 85)? Common errors: 12 (letting 0 go first, or letting a digit repeat) and 16 (both).

Questions the child ticked "not learnt yet" score 0 but tell you what to teach first. They do not count against the child.